FF-T-51D
January 9, 1975
SUPERSEDING
Int. Fed. Spec. FF-T-0051C
November 20, 1968, and
Fed. Spec. FF-T-51A
September 5, 1952

FEDERAL SPECIFICATION

MAPTACK

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers maptacks.

1.2 Classification. The maptacks shall be of the following types, classes, styles, and sizes, as specified (see 6.2):

Type I - Maptacks, Spherical Head.
Dimensions - See Table I.

Type II - Maptacks, Flat Head.
Class A - Round.
Style 1 - Unmarked.
Dimensions - See Table II.

Type III - Maptacks, Marking, Flat Head, Round.
Dimensions - See Table III.

Type IV - Maptacks, Rectangular Head.
Dimensions - See Table IV.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Boxes, Setup.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC; Atlanta, Chicago, Kansas City, MO; Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

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(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks, and the Index of Federal Specifications and Standards, from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, NW., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Standard product. Except for changes necessary to comply with the requirements of this specification, all maptacks furnished under this specification shall be the manufacturer's standard commercial product.

3.2 Material. The heads of the maptacks shall be made of metal, steel, plastic, or glass, as specified (see 6.2). The pins of the maptacks shall be made of commercial low carbon steel.

3.3 Design and construction. The maptacks shall be made of material as specified in 3.2, and the dimensions of the heads and pins shall be as specified in the applicable paragraphs. The pin shall extend from the underside of the head, and shall be concentric with the axis of the head. The length of the pin shall be measured from the underside of the head to the tip. The heads of type II, III, and IV maptacks shall maintain an angle of 90° ± 5° in relation to the pin, when tested as specified in 4.4.1.2. The free end of the pin shall have a tapered and sharp point. The pin shall not become loose, detached, or break off the head under normal usage.

3.3.1 Type I, spherical head. The heads of type I maptacks shall be spherical in shape, and shall be made of metal, plastic, or glass, as specified (see 6.2). The dimensions of the heads and pins shall be as specified in Table I.

TABLE I. Type I

Diameter of heads ± 0.0156"	Length of pins ± 0.0625"	Diameter of pins
0.1719 inch	0.375 inch	0.024 to 0.040 inch
0.25 inch	0.375 inch	0.024 to 0.040 inch
0.25 inch	0.625 inch	0.024 to 0.040 inch
0.3125 inch	1.25 inches	0.024 to 0.040 inch
0.25 inch	0.50 inch	0.0040 to 0.041 inch

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3.3.2 Type II, flat head, class A, round. The heads of type II, class A maptacks shall be flat and round, and shall be made of metal or plastic, as specified (see 6.2). The dimensions of the heads and pins shall be as specified in Table II.

3.3.2.1 Style 1 (unmarked). The heads of style 1 maptacks shall be solid color or a combination of colors (see 6.2).

TABLE II. Type II, class A, style 1

Diameter of heads ± 0.0156"	Length of pins ± 0.0626"	Diameter of pins
0.1875 inch	0.3125 inch	0.024 to 0.040 inch
0.1875 inch	0.625 inch	0.024 to 0.040 inch
0.25 inch	0.375 inch	0.024 to 0.040 inch
0.25 inch	0.625 inch	0.024 to 0.040 inch

3.3.3 Type III, marking, flat head, round. The heads of type III maptacks shall be made of steel and shall be round in shape. The heads shall be covered with plastic or coated with enamel, and shall be capable of being marked with ink and pencil. The heads shall have a maximum convex curvature of 0.75 inch on the 0.25 inch diameter head size, and 2 inches on the 0.50 inch diameter head size indicated in Table III. The dimensions of the heads and pins shall be as specified in Table III.

TABLE III. Type III

Diameter of heads ± 0.0312"	Length of pins ± 0.0625"	Diameter of pins
0.25 inch	0.3125 inch	0.036 to 0.045 inch
0.50 inch	0.4375 inch	0.036 to 0.045 inch

3.3.4 Type IV maptacks, rectangular head. The heads of type IV maptacks shall be made of metal, plastic, or glass, and shall be rectangular in shape. The dimensions of the head and pin shall be as specified in Table IV.

TABLE IV. Type IV

Dimensions of head ± 0.0156"	Length of pin ± 0.0625"	Diameter of pin
0.25 inch wide 0.50 inch long 0.0625 inch thick	0.625 inch	0.020 to 0.030 inch

3.4 Colors. The heads of type I, II, III, and IV maptacks shall be solid or a combination of colors, as specified (see 6.2).

3.5 Finish. All exposed metal surfaces shall have a corrosion resistant finish. The finish on the heads of type III maptacks shall be capable of being marked with ink and pencil, when tested as specified in 4.4.1.1.

3.6 Workmanship. All surfaces of the maptacks shall have a smooth finish and shall be free from sliver, rough, or sharp edges with the exception of the sharp point of the pin. Components shall not be broken, cracked, or malformed. The enamel or plastic coating, when applicable, shall be free from embedded matters and shall show no evidence of peeling, flaking, or blistering.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Size of lot. All maptacks of the same type, class, style, size, and color presented for inspection at one time shall be considered a lot for purposes of examination and test.

4.2.2 Sampling and inspection for acceptance. Sampling and inspection for acceptance shall be performed in accordance with MIL-STD-105, except where otherwise specified.

4.2.3 Sampling for visual and dimensional examination. Boxes of maptacks shall be taken at random from each lot in accordance with MIL-STD-105. The sample unit shall be one box, except that ten (10) maptacks, taken at random from each box, shall be examined as specified in 4.3.1. The Inspection Level shall be S-3 with an Acceptable Quality Level (AQL) of 2.5 percent defective.

4.2.4 Sampling for test. Boxes of maptacks shall be selected at random from each lot in accordance with MIL-STD-105. The sample unit shall be one box, except that ten (10) maptacks, taken at random from each box, shall be tested as specified in 4.4. The Inspection Level shall be S-1 with an AQL of 4.0 percent defective.

4.2.5 Sampling for preparation for delivery requirements. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one shipping container fully prepared for delivery at one time. The Inspection Level shall be S-2 and the AQL shall be 4.0 defects per hundred units.

4.3 Examination.

4.3.1 Visual and dimensional examination. Samples taken in accordance with 4.2.3 shall be examined for defects listed in Table IV. Any maptack having one or more defects shall be rejected. If the number of defective units in the sample exceeds the acceptance number for that sample, the entire lot represented by the sample shall be rejected.

TABLE IV. Classification of defects

Examine	Defects
Standard product	Maptack not standard product (see 3.1).
Material	Not as specified (see 3.2).
Design and construction	Not as specified (see 3.3).
Color	Not as specified (see 3.4).
Finish	Not as specified (see 3.5).
Workmanship	Not in accordance with 3.6.

4.3.2 Examination of preparation for delivery. Samples selected in accordance with 4.2.5 shall be examined for defects listed in Table V to determine that packaging, packing, and marking comply with the requirements of section 5. Defects of closure listed shall be examined on shipping container fully prepared for delivery.

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TABLE V. Classification of preparation for delivery defects

Examine	Defects
Marking (exterior and interior)	Omitted; incorrect; illegible; or improper size, location, sequence, or method of application.
Materials	Any component missing or not as specified. Any component damaged or affecting serviceability.
Workmanship	Inadequate application of components, such as: incomplete closure of container flaps, loose strapping, improper taping, inadequate stapling. Bulging or distortion of containers.
Content	Number per container is more or less than required.

4.4 Test.

4.4.1 Tests for end item. Samples selected in accordance with 4.2.4 shall be subjected to the following tests. Any maptack in the sample failing one or more tests shall be considered defective. If the number of defective maptacks exceeds the acceptance number for the sample, the lot represented by the sample shall be rejected.

4.4.1.1 Writing test (type III). Half of the sample maptacks shall be marked with ink and the other half by pencil. The markings on the maptacks shall be sharp and clear.

4.4.1.2 Head angle test (type II, III, and IV). Type II, III, and IV maptacks shall be inserted in a commercial tackboard at a 60° angle from the surface of the tackboard. Remove maptacks by prying under the edge of the heads, and check for compliance with the requirements stated in 3.3.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packaging. Maptacks of one type, class, style, size, and color only shall be packed in the supplier's commercial box in quantities as follows:

5.1.1.1.1 Type I. In boxes of 50 or 100, as specified (see 6.2).

5.1.1.1.2 Type II, style 1. In boxes of 50 or 100, as specified (see 6.2).

5.1.1.1.3 Type III and IV. In boxes of 25, 50, or 100, as specified (see 6.2).

5.1.1.2 Intermediate packaging. Ten boxes of maptacks of one type, class, style, size, and color only, unit packaged as specified in 5.1.1.1, shall be packaged in a snug-fitting folding, setup, paperboard, or fiberboard box conforming to PPP-B-566, PPP-B-676, or PPP-B-636. Each box shall be securely closed in accordance with the appendix to the applicable box specification.

5.1.2 Level B.

5.1.2.1 Unit packaging.

5.1.2.1.1 Type I, single color. One hundred (100) maptacks of a single type, class, style, size, and color shall be packaged in a snug-fitting paperboard box. Each box shall be securely closed to prevent accidental spillage of contents.

5.1.2.1.2 Type I, mixed colors. Fifteen (15) maptacks each of colors gold, silver, white, orange, blue, green, and ten (10) each of yellow shall be packaged in a paperboard box as specified in 5.1.2.1.1.

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5.1.2.1.3 Type II, style 1. One hundred (100) maptacks shall be packaged in a paperboard box as specified in 5.1.2.1.1.

5.1.2.1.4 Type III and IV. One hundred (100) marking maptacks shall be packaged in a box as specified in 5.1.2.1.1.

5.1.2.2 Intermediate packaging. Ten unit packages of one stock number shall be intermediately packaged in a snug-fitting paperboard or fiberboard box conforming to PPP-B-566 or PPP-B-636. Each box shall be securely closed to prevent accidental spillage of the contents.

5.1.3 Level C. Maptacks shall be packaged to assure adequate protection against damage during shipment from the supply source to the initial destination.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Five hundred (500) unit packages of maptacks of one type, class, style, size, and color, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to PPP-B-636, class weather-resistant, grade V2s. Closure and waterproofing by means of tape and reinforcing shall be in accordance with the appendix of PPP-B-636.

5.2.2 Level B. Maptacks of a single type, class, style, size, and color, packaged as specified in 5.1, shall be packed in quantities as specified (see 6.2) in a snug-fitting box conforming to PPP-B-636, class domestic. Closure of boxes shall be in accordance with method II of the appendix to PPP-B-636.

5.2.3 Level C. Maptacks, in quantities as specified (see 6.2), packaged as specified in 5.1, shall be packed in containers to assure carrier acceptance and safe arrival at destination in compliance with the Uniform Freight Classification Rules or the National Motor Freight Classification Rules.

5.3 Marking. In addition to any special marking required by the procurement documents, interior packages and the shipping containers shall be marked in accordance with Fed. Std. No. 123 for civil agencies and MIL-STD-129 for military agencies as applicable, except that commercial markings on the unit packages will be acceptable.

6. NOTES

6.1 Intended use. Type I, II, and III maptacks covered by this specification are intended for identifying specific points on maps and charts. Type I maptack with the pin diameter of 0.040 to 0.041 inch is intended to designate field artillery battery centers and to accept the vertex of a range reflection protractor. Type IV maptack is intended to designate targets and critical points on the field artillery firing chart.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, class, style, and size required (see 1.2).
(c) Material required for the heads of type I, II, and IV maptacks (see 3.3.1 and 3.3.2).
(d) Specify color(s) and whether solid color or a combination of colors are required (see 3.4).
(e) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(f) Quantity of maptacks per package (see 5.1.1.1.1, 5.1.1.1.2, and 5.1.1.1.3).
(g) Specify special marking, if required (see 5.3).

6.3 Change of classification. This specification includes numerous changes. Type II, classes B, C, and D, and styles 2 and 3 have been deleted. Also, maptacks of various sizes have been deleted. Type IV is a new maptack to designate artillery battery centers.

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